Power Budget

Team Number: 207
Project Name: **Dorm Hub**
Team Member Names: **Aziyah, Dahryl, Chris**
Version: **2**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Microcontroller Board	PIC18f57Q43	+1.8V -5.1V	1	25	25	mA
	Temp/Humid Senor	DHT20	+2.2V-5.5V	1	0.98	0.98	mA
	Debug LED	Red LED	+5.0V	1	10	10	mA
	Bus Pull-Up Resistor	R5,R6	+5.0V	2	1.06	2.13	mA
	Pushbuttons Pull-Up	R2,R3	+5.0V	2	0.5	1	mA

+12V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
							mA
							mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	0	mA
Regulator	N/A						mA
					Total Remaining Current Available on +12V Rail	0	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Microcontroller Board	PIC18f57Q43	+1.8V -5.1V	1	25	25	mA
	Temp/Humid Senor	DHT20	+2.2V-5.5V	1	0.98	0.98	mA
	Debug LED	Red LED	+5.0V	1	10	10	mA
	Bus Pull-Up Resistor	R5,R6	+5.0V	2	1.06	2.13	mA
	Pushbuttons Pull-Up	R2,R3	+5.0V	2	0.5	1	mA
					Subtotal	39.11	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	48.8875	mA
Regulator	+5V Linear Regulator	L7805ABV-DG	+7.0V-35.0	1	150	150	mA
Total Remaining Current Available on +5V Rail						101.1125	mA
+3.3V Power Rail	**Component Name**	**Part Number**	**Supply**	**Qty.**	**Absolute**	**Total**	**Unit**
							mA
						0	mA
			Subtotal			0	mA
			Safety Margin			25%	
			Total Current Required on +3.3V Rail			0	mA
Regulator							mA
Total Remaining Current Available on 3.3V Rail						0	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	Triad WSU120_700	100-240VAC	+12V	700	700	mA
Power Rails Connected to External Power Source 1	+5V Linear Regulator	L7805ABV-DG	+7.0V-35.0	1	150	150	mA
Total Remaining Current Available on External Power Source 1						550	mA

Calculated Battery Life							
	Component Name	**Part Number**	**Supply Voltage Range**		**Capacity (mAh)**	**Required By Regulators**	
	Battery	(full part number)	+12V		500	#REF!	
					Battery Life	#REF!	hours

Notes:

External Supply Voltage

If you have multiple units